PM12 CFD simulator DAC characterization

FIT hardware acceptance bench

10 November 2026

1 Identification and connections

Board: PM12-K7-3A17A1B402212223 (previous FPGA DNA identification). MCU EEPROM serial 1803 was checked in this session. Four CFD simulators occupy physical PM channels 12, 10, 8 and 6. Their analog monitor outputs share one measurement bus. Only one MAX4051 monitor output is enabled at a time. All four monitors are inhibited between transfer curves and on exit. Other GPIO controls are preserved.

Instrument: Keysight B2910BL, serial MY63320299. The monitor bus connects to Force HI/LO; Sense is unconnected. Local (2-wire) voltage measurement is used, with a current source setting of 0 A, 1 microampere source range, 6 V compliance, 20 V measurement range and 1 PLC integration. OUTP is enabled during measurement; automatic output-on is disabled. OUTP OFF/high impedance is restored on exit. The owner explicitly confirmed Force wiring and authorized 0 A/OUTP ON operation.

2 Procedure and scope

The remaining channels 10, 8 and 6 are scanned. DAC B is observed through DRV_OFFSET (monitor 3), DAC C through CFD_OFFSET (monitor 0), and DAC D through CFD_CMP_OFFSET (monitor 1). Phase DAC A is unchanged. The resistor networks produce decreasing spans of approximately 40 mV for B and 20 mV for C, and an increasing span of approximately 1.06 V for D.

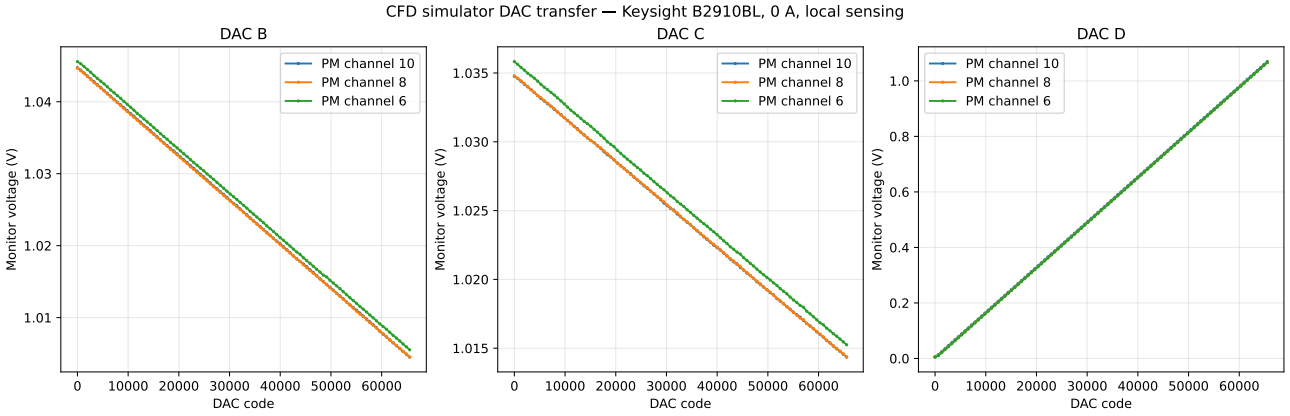
Each DAC is scanned over 101 rounded integer codes spanning 0–65535, with three fresh READ? samples per code and 100 ms settling after each write. Compliance, protection and overflow bits are checked. Raw status words and timestamps are retained; unspecified status bits are ignored per the Keysight SCPI reference, Table 1-2. Each exercised DAC is explicitly set to 0000 on exit. STORED firmware parameters are not treated as active hardware readback.

This is a functional DC transfer scan, not exhaustive 16-bit INL/DNL or a calibrated accuracy certificate. Three repeats estimate short-term scatter. No independent calibrated reference was available; freshness was checked by switching between the 1.3 V rail and DRV_VOCM on channel 12.

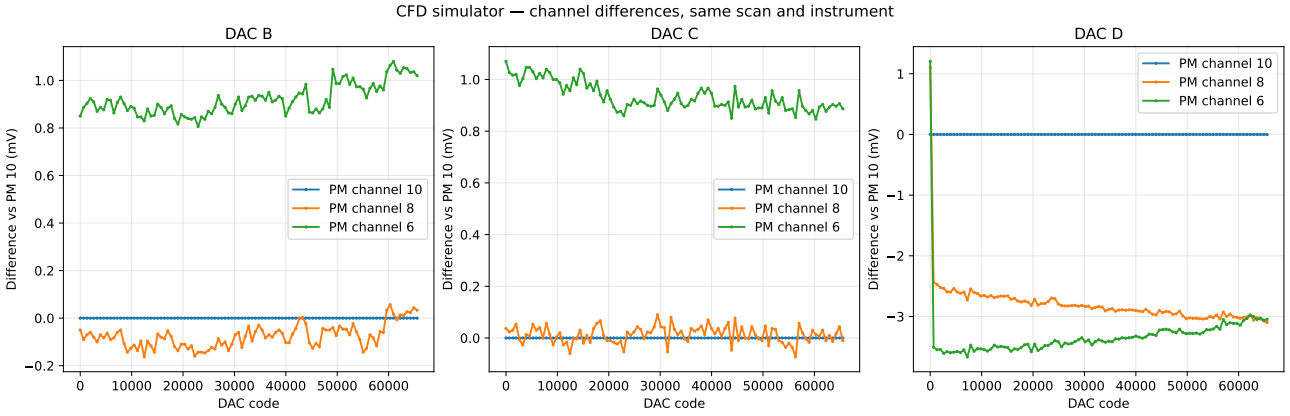
3 Measured endpoint summary

PM ch.	DAC	Code 0 (V)	Code 65535 (V)	Span (V)	Max. s (μV)
10	B	1.044730	1.004507	-0.040223	68.1
8	B	1.044680	1.004540	-0.040140	60.3
6	B	1.045580	1.005527	-0.040053	49.3
10	C	1.034757	1.014360	-0.020397	76.4
8	C	1.034793	1.014350	-0.020443	66.6
6	C	1.035827	1.015247	-0.020580	56.9
10	D	0.004743	1.069480	1.064737	55.1
8	D	0.005847	1.066383	1.060537	69.3
6	D	0.005947	1.066433	1.060487	60.0

4 Transfer curves



The physical channels are overlaid for each DAC.



Differences use channel 10 as reference within the same run.

5 Gain and lower endpoint behavior

All nine curves are monotonic on the sampled grid. Gain is fitted over 5–95% of the code range and compared with the nominal DAC/resistor scaling. The gain deviations below characterize the complete analog chain. The zero-code residual compares the measured minimum-code voltage with the extrapolated interior fit. DAC D shows a distinct lower-end departure; this scan alone does not locate its cause within the DAC and analog stages.

PM ch.	DAC	Interior gain error (%)	Zero-code residual (mV)
10	B	0.226	-0.066
8	B	0.016	0.008
6	B	-0.177	-0.047
10	C	-0.055	-0.051
8	C	-0.047	-0.028
6	C	0.612	0.019
10	D	0.034	1.406
8	D	-0.013	5.096
6	D	0.086	6.261

6 Evidence and final state

Raw samples: `samples.csv`. Statistics: `summary.csv/json`. Commands, I2C readbacks, restoration receipts, instrument identity and final state: `scan-report.json`. Completeness and status checks are in `validation.json`; their PASS is not a calibrated analog acceptance limit. Figures: `plot_smu_dacs.py`; editable chapter: `report_smu_dacs.py`.

Acquisition completed: True.

Final SMU output: 0 (0 means OFF). All nine DAC restoration receipts confirm raw code 0000. Final mux inhibition and disconnected upstream TCA masks are checked before releasing the leases.